REFLEXIVITY IS LOCAL, OR HOLDS WHERE THE DEPTH IS AT LEAST TWO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Localisation of the double dual is the double dual of the localisation. Algebra, Lemma \ref{algebra-lemma-hom-from-finitely-presented}, cited at proof lines 5526-5527, and Lemma \ref{lemma-check-reflexive}, cited at proof lines 5521-5522. Identify the localised double dual and give the easy direction.
- A finite module over a Noetherian domain has torsion free double dual. Lemma \ref{lemma-hom-into-torsion-free}, cited at proof lines 5531-5532. Makes the depth-and-associated-primes criterion applicable to the natural map.
- The criterion for a map to be an isomorphism via depth and associated primes. Lemma \ref{lemma-isom-depth-2-torsion-free}, cited at proof lines 5529-5531. Turns the local hypotheses into the conclusion that the natural map is an isomorphism.

Source reference. Original lemma 0AVA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13678–13678. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0AV1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13669–13669. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0583: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6766–6766. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0AV9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13677–13677. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0AUX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13665–13665. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a Noetherian domain. Let M be a finite R-module. The following are equivalent*

(1) *M is reflexive,*
(2) *for every prime $\mathfrak{p}$ of R one of the following happens*
   (a) *$M_\mathfrak{p}$ is a reflexive $R_\mathfrak{p}$-module, or*
   (b) *$\text{depth}(M_\mathfrak{p}) \geq 2$.*

Proof. If (1) is true, then $M_\mathfrak{p}$ is a reflexive module for all primes of $\mathfrak{p}$ by Lemma 0AV1. Thus (1) $\Rightarrow$ (2). Assume (2). Set $N = \text{Hom}_R(\text{Hom}_R(M, R), R)$ so that

$$N_\mathfrak{p} = \text{Hom}_{R_\mathfrak{p}}(\text{Hom}_{R_\mathfrak{p}}(M_\mathfrak{p}, R_\mathfrak{p}), R_\mathfrak{p})$$

for every prime $\mathfrak{p}$ of R. See Algebra, Lemma 0583. We apply Lemma 0AV9 to the map $j : M \to N$. This is allowed because M is finite and N is torsion free by Lemma 0AUX. In case (2)(a) the map $M_\mathfrak{p} \to N_\mathfrak{p}$ is an isomorphism and in case (2)(b) we have $\text{depth}(M_\mathfrak{p}) \geq 2$. □